Metallic Bonding and Types of Solids

Metals, alloys and the four types of solid

Name _____ Class _____ Date _____

DATA

Melting points: Na 98 °C · Mg 650 °C · Cu 1085 °C · W 3422 °C · NaCl 801 °C · I_2 114 °C

SiO_2 about 1700 °C · diamond above 3500 °C (does not melt at 1 atm)

Explain each property using the particles and the forces between them.

A. Metallic bonding

1. (a) Describe metallic bonding. (b) Use it to explain why metals conduct electricity and why they are malleable.

2. Magnesium (melting point 650 °C) has a higher melting point than sodium (98 °C). Suggest a reason, in terms of the ions and the electron sea.

3. (a) What is an alloy? (b) Explain why bronze is harder than pure copper. (c) A 40.0 g bronze medal is 88.0 % copper by mass. Calculate the mass of copper and of tin.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

B. Types of solid

4. Classify each solid as ionic, molecular, covalent network or metallic, and explain. P: melts at 1085 °C, conducts as a solid. Q: melts at 114 °C, does not conduct. R: melts at 801 °C, conducts when dissolved in water. S: very high melting point, does not conduct at any temperature.

5. Explain why: (a) solid potassium chloride does not conduct, but its solution does (b) iodine has a low melting point even though the I–I bond is covalent.

C. Diamond and graphite

6. Diamond and graphite are both pure carbon. Explain, in terms of structure and bonding, why (a) diamond is very hard (b) graphite conducts electricity (c) graphite is used in pencils.

D. Comparing structures

7. Explain why a metal spoon bends when you hit it, but a crystal of salt shatters.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

E. Challenge

8. Silicon dioxide (SiO_2) melts at about 1700 °C, but carbon dioxide (CO_2) is a gas at room temperature. Both have the formula XO_2. Explain the huge difference.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.